WORKED SOLUTION

STEP I 2003, Question 12

Probability and Statistics • Expectation of Products

STEP I 2003, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q12

The key idea

An expected value is a sum of (value $\times$ probability) over all equally likely outcomes, so both parts come down to listing the ordered pairs of numbers and adding up their products. The double sum $\sum_r \sum_s rs$ factorises as $\left(\sum r\right)^2$, and that is the whole of part (i). In part (ii) the only change is that the pairs with $r = s$ are no longer possible, so we take the same double sum and remove its diagonal.

Solution

Let X be the number on the first ball and Y the number on the second.

Part (i)

Because the first ball is replaced, the second draw is made from the full bag again and does not depend on the first. So for any $r, s \in \{1, \ldots, n\}$,

$$\mathrm{P}(X = r,\ Y = s) = \mathrm{P}(X = r)\,\mathrm{P}(Y = s) = \frac{1}{n} \cdot \frac{1}{n} = \frac{1}{n^2}.$$

There are n^2 such ordered pairs (r, s), each giving product rs, so by the definition of expectation

$$\mathrm{E}(XY) = \sum_{r=1}^{n} \sum_{s=1}^{n} rs\,\mathrm{P}(X = r,\ Y = s) = \frac{1}{n^2} \sum_{r=1}^{n} \sum_{s=1}^{n} rs,$$

which is the expression we were asked to explain.

To simplify it, notice that r does not depend on s, so it comes outside the inner sum:

$$\sum_{r=1}^{n} \sum_{s=1}^{n} rs = \sum_{r=1}^{n} r\left(\sum_{s=1}^{n} s\right) = \left(\sum_{r=1}^{n} r\right)^2 = \tfrac{1}{4}n^2(n+1)^2.$$

Therefore

$$\mathrm{E}(XY) = \frac{(n+1)^2}{4}.$$

(Equivalently, X and Y are independent, so $\mathrm{E}(XY) = \mathrm{E}(X)\,\mathrm{E}(Y) = (\frac{n+1}{2})^2$.)

Part (ii)

We need $n \geq 2$ for two balls to be drawn. Now the first ball is not replaced, so the possible outcomes are the ordered pairs (r, s) with $r \neq s$. For such a pair,

$$\mathrm{P}(X = r,\ Y = s) = \mathrm{P}(X = r)\,\mathrm{P}(Y = s \mid X = r) = \frac{1}{n} \cdot \frac{1}{n-1},$$

since after ball r is removed the second ball is equally likely to be any of the remaining $n - 1$. There are $n(n-1)$ such pairs, all equally likely, and the pairs with $r = s$ have probability 0. Hence

$$\mathrm{E}(XY) = \frac{1}{n(n-1)} \sum_{r \neq s} rs.$$

Ask yourself How do we add up rs over only the pairs with $r \neq s$?

Take the sum over *all* pairs, which we found in part (i), and subtract the diagonal terms $r = s$, which contribute $r \cdot r = r^2$:

$$\sum_{r \neq s} rs = \sum_{r=1}^{n} \sum_{s=1}^{n} rs - \sum_{r=1}^{n} r^2 = \tfrac{1}{4}n^2(n+1)^2 - \tfrac{1}{6}n(n+1)(2n+1).$$

Taking out the common factor $\frac{1}{12}n(n+1)$,

$$\begin{aligned} \sum_{r \neq s} rs &= \tfrac{1}{12}n(n+1)[3n(n+1) - 2(2n+1)] \\ &= \tfrac{1}{12}n(n+1)(3n^2 - n - 2) \\ &= \tfrac{1}{12}n(n+1)(n-1)(3n+2). \end{aligned}$$

Dividing by $n(n-1)$, which is non-zero because $n \geq 2$,

$$\mathrm{E}(XY) = \frac{(n+1)(3n+2)}{12},$$

as required.

Checks

Take $n = 3$. With replacement, the nine products have mean $\frac{1}{9}(1+2+3)^2 = 4$, and $\frac{(3+1)^2}{4} = 4$. Without replacement, the unordered pairs $\{1,2\}, \{1,3\}, \{2,3\}$ give products $2, 3, 6$ with mean $\frac{11}{3}$, and $\frac{4 \cdot 11}{12} = \frac{11}{3}$. For $n = 2$ the only product without replacement is 2, and the formula gives $\frac{3 \cdot 8}{12} = 2$. We also confirmed both formulae by exact enumeration for every n up to 12.

Comparing the two answers is a good sanity check too:

$$\frac{(n+1)(3n+2)}{12} - \frac{(n+1)^2}{4} = -\frac{n+1}{12} < 0.$$

Without replacement the answer is smaller, which makes sense: we have lost the large diagonal products such as $n \cdot n$.

Common slips

- Dividing by n^2 in part (ii). There are only $n(n-1)$ possible ordered pairs, and each has probability $\frac{1}{n(n-1)}$.
- Using unordered pairs for the sum but ordered pairs for the count (or the other way round). Summing over unordered pairs halves both the sum and the count, so the ratio is the same, but mixing the two is out by a factor of 2.
- Writing $\mathrm{E}(XY) = \mathrm{E}(X)\,\mathrm{E}(Y)$ in part (ii). That needs independence, and X and Y are not independent when the ball is not replaced.

Beyond STEP

The difference we found is exactly the covariance: $\mathrm{Cov}(X, Y) = \mathrm{E}(XY) - \mathrm{E}(X)\,\mathrm{E}(Y) = -\frac{n+1}{12}$, and it is negative because a large first number makes a large second number slightly less likely. Covariances like this are what you need to find the variance of a sum when sampling without replacement, which leads to the variance of the hypergeometric distribution and the "finite population correction" used in surveys. Blitzstein and Hwang, *Introduction to Probability*, develops covariance and indicator methods of exactly this kind.

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